

README

Replication Package for “The Geographic Footprint of U.S. Dairy Policy”

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1 OVERVIEW

The code in this package allows one to replicate the results in “The Geographic Footprint of U.S. Dairy Policy” (*Quantitative Economics*) by Hanon, Mérel, and Sumner.

The package contains R code to construct the data files needed to run the main GAMS code from the raw data files, the GAMS code itself, and additional R code that turns the csv files generated by the GAMS code into text files. Each text file corresponds to a table from the main text or the Supplemental Appendix. There are 7 results tables in the main text labeled from Table 1 to Table 7, and additional results tables in the Supplemental Appendix (Tables D.1–D.4 and Table E.1). The package also contains Mathematica code that produces Figure 1 in the main text and the figures in Supplemental Appendix G, as well as a handful of numbers used to substantiate a claim made in Supplemental Appendix G. Figure A.1 in Supplemental Appendix A is generated from a separate R code file.

Finally, the GAMS code produces numbers that are referred to in various sections of the main text and the Supplemental Appendix and that are not part of a table. It also produces numbers that are used to substantiate various claims regarding results not directly reported in a table.

The replication package runs in approximately two minutes.

1.1 DESCRIPTION OF PACKAGE CONTENTS

The top-level directory of this package, titled `Geographic-Footprint-US-Dairy`, contains several subfolders and the following files:

- This README file (in both `.pdf` and `.md` format).
- `README.qmd`, a Quarto file used to generate this README.
- `LICENSE.txt`, which contains the text of the Apache 2.0 license used for the code in this replication package and the Open Data Commons Public Domain Dedication and License v1.0 used for the data in this replication package.
- `Geographic-Footprint-US-Dairy.Rproj`, a file designating this directory as an RStudio Project.
- `.Rprofile`, a file containing code that will be executed when R loads. Specifically, the R script `activate.R` in the `renv` is executed to enable the `{renv}` R package and allow the necessary environment to be replicated. See Section 6 or the [{renv} documentation](#) for more details.
- `renv.lock`, a file with metadata for each R package used in this project. See the [{renv} documentation](#) for more details.

The subfolders contained within the top-level directory include:

- **Data:** Includes raw data in **Raw** subfolder, intermediate data in **Processed** subfolder, **Lookup_Tables** subfolder, and **Other_Sources** subfolder with PDF copies of specific data sources.
- **GAMS CODE:** Contains the main model code to be executed in GAMS Studio. Final data for use in the model are saved in a subfolder called **CSV DATA FILES** during processing.
- **MANUSCRIPT CLAIMS**, **MANUSCRIPT FIGURES**, and **MANUSCRIPT TABLES:** Folders to contain the finished tables, figures, and in-text numerical claims.
- **MATHEMATICA CODE:** Contains the code for Wolfram Mathematica used to produce figures.
- **R CODE:** Contains R code used for processing raw data for use in the main model (**Data-Processing** subfolder) and for processing model results into tables (**Tables** subfolder), figures (**Figures** subfolder), and in-text numerical claims (**Claims** subfolder).
- **References:** Contains **.bib** files for data and R package citations as well as a **.csl** file.
- **renv:** Contains R code and settings files related to **{renv}** package management. When **renv::restore** is run, a **library** subfolder will be created to contain all of the R packages used in this project.

The directory tree below outlines the top two levels of the project directory.

```

../Geographic-Footprint-US-Dairy
├── .Rprofile
├── Data
│   ├── Lookup_Tables
│   ├── Other_Sources
│   ├── Processed
│   └── Raw
├── GAMS CODE
│   ├── CSV DATA FILES
│   ├── FMMO_Model_QE.gms
│   └── FMMO_Model_QE.gsp
├── Geographic-Footprint-US-Dairy.Rproj
├── LICENSE.txt
├── MANUSCRIPT CLAIMS
├── MANUSCRIPT FIGURES
├── MANUSCRIPT TABLES
│   ├── Table_C2.txt
│   ├── Table_C3.txt
│   ├── Table_C4.txt
│   ├── Table_C5.txt
│   └── Table_C6.txt
├── MATHEMATICA CODE
│   ├── FMMOQE.nb
│   └── FMMOQE.wls
├── R CODE
│   ├── Claims
│   ├── Data-Processing
│   ├── Figures
│   └── Tables
└── README.md

```

```
├─ README.pdf
├─ README.qmd
├─ README.rmarkdown
├─ References
│   ├─ data-citations.bib
│   ├─ econometrica.csl
│   └─ grateful-refs.bib
├─ renv
│   ├─ activate.R
│   ├─ library
│   └─ settings.json
├─ renv.lock
└─ run_all.sh
```

2 DATA AVAILABILITY STATEMENTS

All of the data we used in this analysis are public data originally produced by either state or federal government sources. In most cases, we accessed and downloaded the data through online tools. We walk through the steps to access each data source collected this way in the following sections. For a few files, we obtained the raw data directly from the government agencies that produced the data. Table 1 lists the data files in the [Data/Raw](#) folder, all of which are provided in this replication package.

2.1 STATEMENT ABOUT RIGHTS

- ☒ We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.
- ☒ We certify that the authors of the manuscript have documented permission to redistribute/publish the data contained within this replication package. Appropriate permission are documented in the [LICENSE.txt](#) file.

2.2 STATEMENT OF AVAILABILITY

- ☒ All data are publicly available.

Table 1 lists each data file in the [Data/Raw](#) folder and states their availability.

2.3 LICENSE FOR DATA

The data included in this replication package are made available under the Public Domain Dedication and License v1.0 whose full text can be found at: <http://opendatacommons.org/licenses/pddl/1.0/>

See [LICENSE.txt](#) for details.

2.4 AGRICULTURAL MARKETING SERVICE DATA

We used Federal Milk Marketing Order (FMMO) data from the USDA Agricultural Marketing Service (AMS). We accessed and downloaded most of the AMS data through the [Datamart](#) service.

Producer Milk by State of Origin

- Files:

- MilkAndMilkPooledByState2017.csv
- MilkAndMilkPooledByState2019.csv
- Source: USDA Agricultural Marketing Service
- Provided: Yes

These data show the total milk produced in each state and the quantity of milk originating in each state that was pooled in each FMMO. These files were provided to the authors by AMS in `.xlsx` format. The files included in this package were simply saved in `.csv` format but are otherwise unedited.

Producer Receipts and Component Tests

- Files:
 - Datamart-Export_FM_Producer_Receipts_2017.csv
 - Datamart-Export_FM_Producer_Components-Butterfat_2017.csv
 - Datamart-Export_FM_Producer_Components-Protein_2017.csv
 - Datamart-Export_FM_Producer_Components-Other Solids_2017.csv

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Producer Milk Components”. Under “Sub Report Type to Include”, check the boxes for Receipts, Butterfat, Protein, and Other Solids (alternatively, uncheck the boxes for NFS and SomCell). Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2017a)).

Class I Utilization of Producer Milk

- Files:
 - Datamart-Export_FM_ClassI_Util-Butterfat_2017.csv
 - Datamart-Export_FM_ClassI_Util-Milk_2017.csv
 - Datamart-Export_FM_ClassI_Util-NFS_2017.csv
 - Datamart-Export_FM_ClassI_Util-Utilization_2017.csv

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class I Utilization of Producer Milk”. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2017b)).

Class II Utilization of Producer Milk

- Files:
 - Datamart-Export_FM_ClassII_Util-Butterfat_2017.csv
 - Datamart-Export_FM_ClassII_Util-NFS_2017.csv
 - Datamart-Export_FM_ClassII_Util-Utilization_2017.csv

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class II Utilization of Producer Milk”. Under “Sub Report Type to Include”, uncheck the box for Milk. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2017c)).

Class III Utilization of Producer Milk

- Files:
 - Datamart-Export_FM_ClassIII_Util-Butterfat_2017.csv
 - Datamart-Export_FM_ClassIII_Util-Other Solids_2017.csv

- [Datamart-Export_FM_ClassIII_Util-Protein_2017.csv](#)
- [Datamart-Export_FM_ClassIII_Util-Utilization_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class III Utilization of Producer Milk”. Under “Sub Report Type to Include”, uncheck the box for Milk. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2017d)).

Class IV Utilization of Producer Milk

- Files:
 - [Datamart-Export_FM_ClassIV_Util-Butterfat_2017.csv](#)
 - [Datamart-Export_FM_ClassIV_Util-NFS_2017.csv](#)
 - [Datamart-Export_FM_ClassIV_Util-Utilization_2017.csv](#)

From Datamart homepage select “FMMOS” from the Commodity menu. Click “Utilization Reports” and “Class IV Utilization of Producer Milk”. Under “Sub Report Type to Include”, uncheck the box for Milk. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2017e)).

Advanced Prices

- Files:
 - [Datamart-Export_DY_Advanced_Prices_2017.csv](#)

From Datamart homepage select “Dairy” from the Commodity menu. Click “All Dairy” and “(DY_AD101) National Dairy Product Prices - Advanced”. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Uncheck all fields on the next screen and select “Base Class 1 Price”, “Base Skim Milk Class 1 Price”, and “Advanced Butterfat Factor”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2017f)).

Class Prices

- Files:
 - [Datamart-Export_DY_Class_Prices_2017.csv](#)

From Datamart homepage select “Dairy” from the Commodity menu. Click “All Dairy” and “(DY_CL102) National Dairy Product Prices - Monthly”. Under Report Year, switch the dropdown menu from “All” to “Equal To” and enter 2017 in the box that appears. Click “Continue”. Uncheck all fields on the next screen and select “Class 2 Price”, “Class 2 Butterfat Price”, “Class 3 Price”, “Class 3 Skim Milk Price”, “Class 4 Price”, “Class 4 Skim Milk Price”, “Butterfat Price”, “Nonfat Solids Price”, “Protein Price”, and “Other Solids Price”. Select CSV as the file format and click “Generate Report” (Agricultural Marketing Service (2018)).

2.5 MILK POOLING DATA FROM STATE SOURCES

In addition to the FMMO data from AMS, we used data from two state milk marketing orders, the California Milk Marketing Order and the Western New York Milk Marketing Area. In both cases, we accessed PDF reports published by the respective state departments of agriculture and generated CSV files with the relevant data. The original PDF reports and the CSV files are both available in this replication package.

California Milk Pooling Data

- Files:

- [California_Dairy_Statistics_Annual_2017.pdf](#)
- [California_Statistics_2017.csv](#)
- [California_Monthly_ClassI_2017.csv](#)
- Source: California Department of Food and Agriculture
- Original URL: https://www.cdfa.ca.gov/dairy/pdf/Annual/2017/2017_Statistics_Annual.pdf
- Archived URL: https://web.archive.org/web/20181130220414/https://www.cdfa.ca.gov/dairy/pdf/Annual/2017/2017_Statistics_Annual.pdf

We obtained data on milk volumes and utilization of components by class in the California State Milk Marketing Order from the California Dairy Statistics Annual report for 2017 published by the California Department of Food and Agriculture (California Department of Food and Agriculture (2017). We created two CSV files with data from a PDF copy of the report, which itself is included in the [Data/Other_Sources](#) folder. The report is no longer accessible through the CDFA website, but can be accessed through the *Internet Archive*. An archival URL is included.

Western New York Milk Pooling Data

- Files:
 - [Western_New_York_Annual_Statistical_Report_2017.pdf](#)
 - [Western_New_York_Data_2017.csv](#)
- Source: New York Department of Agriculture and Markets
- Original URL: <https://agriculture.ny.gov/system/files/documents/2019/06/WNYAnnual2017.pdf>

We obtained data on milk volumes and utilization of components by class in the Western New York Milk Marketing Area from the Western New York Milk Marketing Area Annual Statistical Report 2017 published by the New York Department of Agriculture and Markets (New York Department of Agriculture and Markets (2017). We create a CSV file with data from a PDF copy of the report, which itself is included in the [Data/Other_Sources](#) folder and can be accessed through the [New York Department of Agriculture and Markets website](#).

2.6 NATIONAL AGRICULTURAL STATISTICS SERVICE DATA

The National Agricultural Statistics Service (NASS) data we used fall into two general categories: Census data and Survey data. We used Census data from the 2017 Census of Agriculture (National Agricultural Statistics Service (2019). When specific data series were unavailable in the 2017 Census, or when data from years other than 2017 were needed, we used data from NASS survey reports. Those survey reports are cited individually below. In all cases, we used the [NASS Quick Stats](#) service to download data. The steps to access each raw data file are outlined below.

Milk Production

- Files:
 - [NASS_Milk_Production_2017.csv](#)

We used survey data on state-level milk production volumes from the *Milk Production* report (National Agricultural Statistics Service (2018a). To download these data from Quick Stats, select “MILK” from the Commodity box; select “PRODUCTION” from the Category box; select “MILK - PRODUCTION, MEASURED IN LB” from the Data Item box; select “STATE” from the Geographic Level box; select 2017 from the Year box; select “ANNUAL” from the Period Type box.

Dairy Products

- Files:
 - [NASS_Dairy_Products_2017.csv](#)

We used survey data on state and multi-state regional dairy product production volumes from the *Dairy Products Annual Summary* report (National Agricultural Statistics Service (2018b)). To download these data from Quick Stats, select “SURVEY” from the Program box; select “ANIMALS & PRODUCTS” from the Sector box; select “DAIRY” from the Group box; select “BUTTER”, “CHEESE”, “CREAM”, “ICE CREAM”, “MILK”, “SHERBET”, “WHEY”, and “YOGURT” from the Commodity box; select “PRODUCTION” from the Category box; select all of the following from the Data Item box:

- BUTTER - PRODUCTION, MEASURED IN LB
- CHEESE - PRODUCTION, MEASURED IN LB
- CHEESE, COTTAGE, CREAMED, FAT CONTENT GE 4 PCT - PRODUCTION, MEASURED IN LB
- CHEESE, COTTAGE, LOWFAT, FAT CONTENT LT 4 PCT - PRODUCTION, MEASURED IN LB
- CREAM, SOUR - PRODUCTION, MEASURED IN LB
- ICE CREAM, LOWFAT - PRODUCTION, MEASURED IN GALLONS
- ICE CREAM, MIX, LOWFAT - PRODUCTION, MEASURED IN GALLONS
- ICE CREAM, MIX, REGULAR - PRODUCTION, MEASURED IN GALLONS
- ICE CREAM, REGULAR, HARD - PRODUCTION, MEASURED IN GALLONS
- MILK, DRY, NONFAT, HUMAN - PRODUCTION, MEASURED IN LB
- MILK, SKIM, UNSWEETENED, CONDENSED, BULK - PRODUCTION, MEASURED IN LB
- SHERBET, HARD - PRODUCTION, MEASURED IN GALLONS
- SHERBET, MIX - PRODUCTION, MEASURED IN GALLONS
- WHEY, DRY, HUMAN - PRODUCTION, MEASURED IN LB
- YOGURT, MIX, FROZEN - PRODUCTION, MEASURED IN GALLONS
- YOGURT, PLAIN & FLAVORED - PRODUCTION, MEASURED IN LB

Select “REGION: MULTI-STATE” and “STATE” from the Geographic Level box; select 2017 from the Year box; select “ANNUAL” from the Period Type box.

Total Cropland and Harvested Cropland Acreage

- Files:
 - [NASS_All_Cropland_2017.csv](#)
 - [NASS_All_Cropland_Harvested_2017.csv](#)

We used 2017 Census data on total cropland acreage and harvested cropland acreage (National Agricultural Statistics Service (2019)). To download these data from Quick Stats, for both files select “CENSUS” from the Program box; select “AG LAND” from the Commodity box; select “AREA” from the Category box.

For the file [NASS_All_Cropland_2017.csv](#), select “AG LAND, CROPLAND - ACRES” and “AG LAND, CROPLAND, PASTURED ONLY - ACRES” from the Data Item box. For the file [NASS_All_Cropland_Harvested_2017.csv](#), select “AG LAND, CROPLAND, HARVESTED - ACRES” from the Data Item box. For both files, select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Feed Crop Acreage

- Files:
 - [NASS_Census_Feed_Crop_Acreage_2017.csv](#)

We used 2017 Census data on feed crop acres harvested (National Agricultural Statistics Service (2019)). To download these data from Quick Stats, select “CENSUS” from the Program box; select “FIELD CROPS” from the Group box; select “BARLEY”, “BUCKWHEAT”, “CANOLA”, “CORN”, “COTTON”, “FLAXSEED”, “HAY”, “HAYLAGE”, “MILLET”, “OATS”,

“RICE”, “RYE”, “SORGHUM”, “SOYBEANS”, and “WHEAT” from the Commodity box; select “AREA HARVESTED” from the Category box; select all of the following from the Data Item box:

- BARLEY - ACRES HARVESTED
- BUCKWHEAT - ACRES HARVESTED
- CANOLA - ACRES HARVESTED
- CORN, GRAIN - ACRES HARVESTED
- CORN, SILAGE - ACRES HARVESTED
- COTTON - ACRES HARVESTED
- FLAXSEED - ACRES HARVESTED
- HAY - ACRES HARVESTED
- HAYLAGE - ACRES HARVESTED
- MILLET, PROSO - ACRES HARVESTED
- OATS - ACRES HARVESTED
- RICE - ACRES HARVESTED
- RYE - ACRES HARVESTED
- SORGHUM, GRAIN - ACRES HARVESTED
- SORGHUM, SILAGE - ACRES HARVESTED
- SOYBEANS - ACRES HARVESTED
- WHEAT - ACRES HARVESTED

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Other Crop Acreage

- Files:
 - [NASS_Census_Other_Crop_Acreage_2017.csv](#)

We used 2017 Census data on other crop acres harvested (National Agricultural Statistics Service (2019)). To download these data from Quick Stats, select “CENSUS” from the Program box; select “FIELD CROPS”, “FRUIT & TREE NUTS”, and “VEGETABLES” from the Group box; select “BEANS”, “BERRY TOTALS”, “CHICKPEAS”, “GRASSES & LEGUMES TOTALS”, “HOPS”, “LENTILS”, “MINT”, “ORCHARDS”, “PEANUTS”, “PEAS”, “POPCORN”, “SAFFLOWER”, “SUGAR-BEETS”, “SUGARCANE”, “SUNFLOWER”, “TOBACCO”, and “VEGETABLE TOTALS” from the Commodity box; select “AREA BEARING & NON-BEARING”, “AREA GROWN”, and “AREA HARVESTED” from the Category box; select all of the following from the Data Item box:

- BEANS, DRY EDIBLE, (EXCL CHICKPEAS & LIMA) - ACRES HARVESTED
- BEANS, DRY EDIBLE, LIMA - ACRES HARVESTED
- BERRY TOTALS - ACRES GROWN
- CHICKPEAS - ACRES HARVESTED
- GRASSES & LEGUMES TOTALS, SEED - ACRES HARVESTED
- HOPS - ACRES HARVESTED
- LENTILS - ACRES HARVESTED
- MINT, OIL - ACRES HARVESTED
- ORCHARDS - ACRES BEARING & NON-BEARING
- PEANUTS - ACRES HARVESTED
- PEAS, DRY EDIBLE - ACRES HARVESTED
- POPCORN, SHELLS - ACRES HARVESTED
- SAFFLOWER - ACRES HARVESTED

- SUGARBEETS - ACRES HARVESTED
- SUGARCANE, SUGAR - ACRES HARVESTED
- SUNFLOWER - ACRES HARVESTED
- TOBACCO - ACRES HARVESTED
- VEGETABLE TOTALS, IN THE OPEN - ACRES HARVESTED

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Crop Sales

- Files:
 - [NASS_Total_Crop_Sales_2017.csv](#)
 - [NASS_Crop_Sales_2017.csv](#)

We used 2017 Census data on total crop sales and individual crop sales (National Agricultural Statistics Service (2019). Four commodities do not have sales reported in the Census of Agriculture, and therefore we used survey data on the value of production for these commodities: hay, peanuts, sugarbeets, and sugarcane. Quick Stats contains records for “PRODUCTION, MEASURED IN \$” that are a combination of production volumes reported in the *Crop Production Annual Summary* report and prices received reported in the *Agricultural Prices* report (National Agricultural Statistics Service (2018c); National Agricultural Statistics Service (2018d).

To download the data in the file [NASS_Total_Crop_Sales_2017.csv](#) from Quick Stats, select “CENSUS” from the Program box; select “CROP TOTALS” from the Group box; select “CROP TOTALS” from the Commodity box; select “SALES” from the Category box; select “CROP TOTALS - SALES, MEASURED IN \$” from the Data Item box; select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

To download the data in the file [NASS_Crop_Sales_2017.csv](#) from Quick Stats, select “FIELD CROPS” from the Group box; select “BARLEY”, “CORN”, “COTTON”, “FRUIT & TREE NUT TOTALS”, “HAY”, “PEANUTS”, “RICE”, “SORGHUM”, “SOYBEANS”, “SUGARBEETS”, “SUGARCANE”, “TOBACCO”, “VEGETABLE TOTALS”, and “WHEAT” from the Commodity box; select “PRODUCTION” and “SALES” from the Category box; select all of the following from the Data Item box:

- BARLEY - SALES, MEASURED IN \$
- CORN - SALES, MEASURED IN \$
- COTTON, LINT & SEED - SALES, MEASURED IN \$
- FRUIT & TREE NUT TOTALS - SALES, MEASURED IN \$
- GRAIN, OTHER - SALES, MEASURED IN \$
- HAY - PRODUCTION, MEASURED IN \$
- PEANUTS - PRODUCTION, MEASURED IN \$
- RICE - SALES, MEASURED IN \$
- SORGHUM - SALES, MEASURED IN \$
- SOYBEANS - SALES, MEASURED IN \$
- SUGARBEETS - PRODUCTION, MEASURED IN \$
- SUGARCANE, SUGAR & SEED - PRODUCTION, MEASURED IN \$
- TOBACCO - SALES, MEASURED IN \$
- VEGETABLE TOTALS, INCL SEEDS & TRANSPLANTS, IN THE OPEN - SALES, MEASURED IN \$
- WHEAT - SALES, MEASURED IN \$

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box.

Silage and Haylage Production Value

- Files:
 - [NASS_Silage_Data_2017.csv](#)
 - [NASS_Haylage_Data_2017.csv](#)
 - [NASS_Silage_Data_1519.csv](#)
 - [NASS_Haylage_Data_1519.csv](#)

We used survey data on silage and haylage production quantities from the 2017 Census and prices received from the *Agricultural Prices* report to compute silage and haylage production values for 2017 (National Agricultural Statistics Service (2019); National Agricultural Statistics Service (2018d)). We also used a range of silage and haylage data from 2015-2019 for the regression analysis described in Appendix C. These data came from the *Crop Production Annual Summary* report and *Agricultural Prices* report. (National Agricultural Statistics Service (2015a); National Agricultural Statistics Service (2015b)).

To download the data in the file [NASS_Silage_Data_2017.csv](#) from Quick Stats, select “FIELD CROPS” from the Group box; select “CORN” and “SORGHUM” from the Commodity box; select “PRICE RECEIVED” and “PRODUCTION” from the Category box; select “CORN, GRAIN - PRICE RECEIVED, MEASURED IN \$ / BU”, “CORN, SILAGE - PRODUCTION, MEASURED IN TONS”, and “SORGHUM, SILAGE - PRODUCTION, MEASURED IN TONS” from the Data Item box; select “STATE” from the Geographic Level box; select 2017 from the Year box; select “ANNUAL” from the Period type box. *Note:* this search will also return survey data for corn and sorghum silage production quantities, but we did not use these data in the analysis.

Downloading the data in the file [NASS_Haylage_Data_2017.csv](#) is nearly identical to the above steps. Instead of selecting “CORN” and “SORGHUM” in the Commodity box, select “HAY” and “HAYLAGE”. In the Data Item box, select “HAY - PRICE RECEIVED, MEASURED IN \$ / TON” and “HAYLAGE - PRODUCTION, MEASURED IN TONS”. As with the silage data, this will return survey data for hay production quantities, but we only used the Census data.

For the data in the files [NASS_Silage_Data_1519.csv](#) and [NASS_Haylage_Data_1519.csv](#), first select “SURVEY” in the Program box, then follow all of the same steps outlined above through the Geographic Level box. Then, select 2015-2019 in the Year box and “ANNUAL” in the Period Type box.

Livestock Inventories

- Files:
 - [NASS_Livestock_Data_1519.csv](#)

We used a combination of livestock inventory data from the 2017 Census and survey data from the *Cattle* report (National Agricultural Statistics Service (2019), National Agricultural Statistics Service ((2015))). For the regression analysis of silage use by dairy cattle in the [4_Silage_Regression.R](#) script, we use the survey data from the *Cattle* report for 2015 through 2019. That way the data are all from a consistent source. The 2017 Census data and survey data on the number of milk replacement heifers in 2017 are used in the [6_Process_Feed_Use_Data.R](#) script.

To download these data from Quick Stats, *do not* make a selection from the Program box; select “ANIMALS & PRODUCTS” from the Sector box; select “LIVESTOCK”, “POULTRY”, and “SPECIALTY” from the Group box; select “CATTLE”, “CHICKENS”, “EQUINE”, “GOATS”, “HOGS”, “SHEEP”, and “TURKEYS” from the Commodity box; select “INVENTORY” from the Category box; select all of the following from the Data Item box:

- CATTLE, COWS, MILK - INVENTORY
- CATTLE, HEIFERS, GE 500 LBS, MILK REPLACEMENT - INVENTORY
- CATTLE, INCL CALVES - INVENTORY
- CATTLE, ON FEED - INVENTORY

- CHICKENS, BROILERS - INVENTORY
- CHICKENS, LAYERS - INVENTORY
- EQUINE, HORSE & PONIES - INVENTORY
- GOATS - INVENTORY
- HOGS - INVENTORY
- SHEEP, INCL LAMBS - INVENTORY
- TURKEYS - INVENTORY

Select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2015 through 2019 from the Year box; select “POINT IN TIME” from the Period Type box; and select “FIRST OF JAN” and “END OF DEC” from the Period box.

Note: This search will also return survey data for sheep inventories from 2015 to 2019, but these data are not used in the analysis.

Cropland Acreage and Rents

- Files:
 - [CroplandAcresState.csv](#)
 - [CroplandRentState.csv](#)

We used cropland acreage data from the 2017 Census and cropland rent data from the NASS *Cash Rents* survey from 2015 through 2019 (National Agricultural Statistics Service (2019); National Agricultural Statistics Service (2015c).

To download the data in the file [CroplandAcresState.csv](#) from Quick Stats, select “CENSUS” from the Program box; select “ECONOMICS” from the Sector box; select “FARMS & LAND & ASSETS” from the Group box; select “AG LAND” from the Commodity box; select “AG LAND, CROPLAND – ACRES” from the Data Item box; select “TOTAL” from the Domain box; select “STATE” from the Geographic Level box; select 2017 from the Year box. Then, we select the columns named Year, State, State ANSI, and Value. The table is ordered by increasing value of the State ANSI variable.

To download the data in the file [CroplandRentState.csv](#) from Quick Stats, select “SURVEY” from the Program box; select “ECONOMICS” from the Sector box; select “EXPENSES” from the Group box; select “RENT” from the Commodity box; select “RENT, CASH, CROPLAND – EXPENSE, MEASURED IN \$/ACRE” from the Data Item box; select “STATE” from the Geographic Level box; select 2010-2023 from the Year box. Then, we select the columns named Year, State, State ANSI, and Value. The table is ordered by increasing value of the State ANSI variable. *Note:* this search returns data for 2010-2023, but we only used data from 2015-2019 in the analysis.

2.7 LOOKUP TABLES

The data include several lookup tables that were created by the authors to facilitate matching values between different data sets. For example, matching states to FMMO regions. The lookup tables are located in the “Lookup Tables” subfolder within the “Data” folder.

- [State_Region_Match_Table.csv](#)
 - Source: Created by the Authors
 - Format: .csv
 - Provided: Yes
 - Notes: Federal Milk Marketing Order regions were assigned to each state based primarily on geography and the quantity of milk pooled. In most cases, a given state is clearly geographically aligned with a FMMO marketing area.

In cases where a state is either split between to FMMO marketing areas or is partially covered by a marketing area and partially unregulated, a determination was made about which region to assign.

3 DATASET LIST

Table 1: List of Files in Data/Raw Folder

File Path	Size	Source	Pro- vided
Data/Raw/California_Monthly_ClassI_2017.csv	179	CDFA	TRUE
Data/Raw/California_Statistics_2017.csv	371	CDFA	TRUE
Data/Raw/Census_Dairy_Product_Exports_2017.csv	74.23K	U.S. Census Bureau	TRUE
Data/Raw/Census_Dairy_Product_Imports_2017.csv	43.16K	U.S. Census Bureau	TRUE
Data/Raw/Census_Population_2017.csv	1.66K	U.S. Census Bureau	TRUE
Data/Raw/CroplandAcresState.csv	1.5K	USDA-NASS	TRUE
Data/Raw/CroplandRentState.csv	13.9K	USDA-NASS	TRUE
Data/Raw/Datamart-Export_DY_Advanced_Prices_2017.csv	573	USDA-AMS	TRUE
Data/Raw/Datamart-Export_DY_Class_Prices_2017.csv	1.18K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Butterfat_2017.csv	1.1K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Other Solids_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Protein_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIII_Util-Utilization_2017.csv	1.21K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassII_Util-Butterfat_2017.csv	1.16K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassII_Util-NFS_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassII_Util-Utilization_2017.csv	1.2K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIV_Util-Butterfat_2017.csv	1.11K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIV_Util-NFS_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassIV_Util-Utilization_2017.csv	1.2K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-Butterfat_2017.csv	1.1K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-Milk_2017.csv	1.05K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-NFS_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_ClassI_Util-Utilization_2017.csv	1.24K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Components-Butterfat_2017.csv	1.1K	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Components-Other Solids_2017.csv	783	USDA-AMS	TRUE

File Path	Size	Source	Pro- vided
Data/Raw/Datamart-Export_FM_Producer_Components-Protein_2017.csv	783	USDA-AMS	TRUE
Data/Raw/Datamart-Export_FM_Producer_Receipts_2017.csv	1.29K	USDA-AMS	TRUE
Data/Raw/ERS_Component_Conversions.csv	323	USDA-ERS	TRUE
Data/Raw/ERS_FCAU_Data.csv	9.66K	USDA-ERS	TRUE
Data/Raw/FAF_Shipments_All_2017.csv	578.45K	U.S. Census Bu- reau	TRUE
Data/Raw/feed-grains-yearbook-all-years.csv	16.22M	USDA-ERS	TRUE
Data/Raw/GATS_Crop_Exports_1519.csv	2.54K	USDA-FAS	TRUE
Data/Raw/MilkAndMilkPooledByState2017.csv	4.25K	USDA-AMS	TRUE
Data/Raw/MilkAndMilkPooledByState2019.csv	4.46K	USDA-AMS	TRUE
Data/Raw/NASS_All_Cropland_2017.csv	12.36K	USDA-NASS	TRUE
Data/Raw/NASS_All_Cropland_Harvested_2017.csv	6.48K	USDA-NASS	TRUE
Data/Raw/NASS_Census_Feed_Crop_Acreage_2017.csv	62.2K	USDA-NASS	TRUE
Data/Raw/NASS_Census_Other_Crop_Acreage_2017.csv	49.77K	USDA-NASS	TRUE
Data/Raw/NASS_Crop_Sales_2017.csv	58.6K	USDA-NASS	TRUE
Data/Raw/NASS_Dairy_Products_2017.csv	19.78K	USDA-NASS	TRUE
Data/Raw/NASS_Haylage_Data_1519.csv	41.79K	USDA-NASS	TRUE
Data/Raw/NASS_Haylage_Data_2017.csv	12.45K	USDA-NASS	TRUE
Data/Raw/NASS_Livestock_Data_1519.csv	195.55K	USDA-NASS	TRUE
Data/Raw/NASS_Milk_Production_2017.csv	6.22K	USDA-NASS	TRUE
Data/Raw/NASS_Silage_Data_1519.csv	69.22K	USDA-NASS	TRUE
Data/Raw/NASS_Silage_Data_2017.csv	17.72K	USDA-NASS	TRUE
Data/Raw/NASS_Total_Crop_Sales_2017.csv	6.76K	USDA-NASS	TRUE
Data/Raw/OilCropsAllTables.csv	7.65M	USDA-ERS	TRUE
Data/Raw/selected-dairy-products-percapita.csv	69.9K	USDA-ERS	TRUE
Data/Raw/Special Tab 10.txt	148.8M	U.S. Census Bu- reau	TRUE
Data/Raw/us-bioenergy-statistics.csv	3.32M	USDA-ERS	TRUE
Data/Raw/Western_New_York_Data_2017.csv	175	NYDAM	TRUE

Table 2: List of Files in Data/Lookup_Tables Folder

File Path	Size	Source	Provided
Data/Lookup_Tables/Census_Regions_Divisions_States.csv	2.43K	Created by the authors	TRUE
Data/Lookup_Tables/Class1_Differentials.csv	206	Created by the authors	TRUE
Data/Lookup_Tables/Crop_Groups.csv	602	Created by the authors	TRUE
Data/Lookup_Tables/HSCode_Class.csv	530	Created by the authors	TRUE
Data/Lookup_Tables/HSCode_Crops.csv	528	Created by the authors	TRUE
Data/Lookup_Tables/NASS_Dairy_Product_Match_Table.csv	607	Created by the authors	TRUE
Data/Lookup_Tables/NASS_Livestock_Match_Table.csv	494	Created by the authors	TRUE
Data/Lookup_Tables/NASS_Regions_Match_Table.csv	891	Created by the authors	TRUE
Data/Lookup_Tables/State_Region_Match_Table.csv	1.26K	Created by the authors	TRUE

Table 3: List of Files in Data/Other_Sources Folder

File Path	Size	Source	Provided
Data/Other_Sources/California_Dairy_Statistics_Annual_2017.pdf	5.58M	CDFA	TRUE
Data/Other_Sources/Western_New_York_Annual_Statistical_Report_2017.pdf	538.42K	NY-DAM	TRUE

4 COMPUTATIONAL REQUIREMENTS

We used R version 4.5.2 and the packages listed in Table 4 running in RStudio version 2026.7.1+147 to process the raw data contained in [Data/Raw](#) for analysis (R Core Team (2025); Posit team (2026). We used GAMS Studio version 1.25.5 and the [CONOPT3](#) solver for the main model. We used R to process the results from the main model to produce the tables in the manuscript and two figures in the supplemental material. Additionally, we used Wolfram Mathematica version 12.2.0.0 to produce the figures in the manuscript and additional figures in the supplemental material.

List of Necessary Software:

- R 4.5.2
- RStudio 2026.07.1+147
- GAMS Studio Version 1.25.5
 - [CONOPT3](#) solver
- Wolfram Mathematica Version Number 12.2.0.0

Table 4: R Packages Used

Package	Version	Citation
base	4.5.2	R Core Team ((2025)
fs	2.1.0	Hester, Wickham, and Csárdi ((2026)
grateful	0.3.0	Rodriguez-Sanchez and Jackson ((2025)
here	1.0.2	Müller ((2025)

Package	Version	Citation
knitr	1.51	Xie ((2014); Xie ((2015); Xie ((2025)
plyr	1.8.9	Wickham ((2011)
renv	1.2.4	Ushey and Wickham ((2026)
rmarkdown	2.31	Xie, Allaire, and Golemund ((2018); Xie, Dervieux, and Riederer ((2020); Allaire et al. ((2026)
sf	1.1.2	Pebesma ((2018); Pebesma and Bivand ((2023)
stargazer	5.2.3	Hlavac ((2022)
tidyverse	2.0.0	Wickham et al. ((2019)

4.1 MEMORY, RUNTIME, AND STORAGE REQUIREMENTS

Summary

Approximate time needed to reproduce the analyses on a standard 2026 desktop machine:

Approximate storage space needed: 638 MB

Details

The data processing portions of the code were last run on a 6-core AMD-based desktop with 32 GB of RAM and 475 GB storage (45.8 GB free) running Windows 11 version 25H2.

The main model code was last run on a 14-inch 2021 Apple Macbook Pro with an Apple M1 chip and 32 GB of RAM running Mac OS Sonoma 14.7.

5 DESCRIPTION OF CODE SCRIPT FILES

5.1 LICENSE FOR CODE

The code included in this replication package is licensed under the Apache 2.0 license. A copy of the license may be obtained at <https://www.apache.org/licenses/LICENSE-2.0>. See [LICENSE.txt](#) for more information.

6 INSTRUCTIONS TO REPLICATORS

7 LIST OF TABLES AND FIGURES AND ASSOCIATED CODE FILES

Table 5: Code Files to Generate Tables and Figures in Manuscript

Figure/Table	Script File	Output File
Table 1	TablesQE.R	table1.txt
Table 2	TablesQE.R	table2.txt
Table 3	TablesQE.R	table3.txt
Table 4	TablesQE.R	table4.txt
Table 5	TablesQE.R	table5.txt
Table 6	TablesQE.R	table6.txt

Figure/Table	Script File	Output File
Table 7	TablesQE.R	table7.txt
Table C.2	3_Process_Component_Values_Volumes.R	Table_C2.txt
Table C.3	6_Process_Feed_Use_Data.R	Table_C3.txt
Table C.4	6_Process_Feed_Use_Data.R	Table_C4.txt
Table C.5	5_Silage_Regression.R	Table_C5.txt
Table C.6	5_Silage_Regression.R	Table_C5.txt
Table D.1	TablesQE.R	tabled1.txt
Table D.2	TablesQE.R	tabled2.txt
Table D.3	TablesQE.R	tabled3.txt
Table D.4	TablesQE.R	tabled4.txt
Table E.1	TablesQE.R	tablee1.txt
Figure 1	FMMOQE.wls	figure1.png
Figure A.1	MapsFMMOQE.R	fmmoregion.png, fmmoapproximations.png
Figure G.1	FMMOQE.wls	figureg1.png
Figure G.2	FMMOQE.wls	figureg2.png
Figure G.3	FMMOQE.wls	figureg3.png
Figure G.4	FMMOQE.wls	figureg4.png
Figure G.5	FMMOQE.wls	figureg5.png

REFERENCES

- AGRICULTURAL MARKETING SERVICE. (2017f): “Advanced Prices and Pricing Factors,”https://mpr.datamart.ams.usda.gov/menu.do?path=Products\Dairy\All%20Dairy\DY_CL102%20National%20Dairy%20Products%20Prices%20-%20Monthly, Accessed 08/13/2026.
- AGRICULTURAL MARKETING SERVICE. (2017a): “Producer Milk Components,”<https://mpr.datamart.ams.usda.gov/menu.do?path=Products\FMMOS%20Utilization%20Reports\Producer%20Milk%20Components>, Accessed 08/10/2026.
- AGRICULTURAL MARKETING SERVICE. (2017b): “Class I Utilization of Producer Milk,”<https://mpr.datamart.ams.usda.gov/menu.do?path=\Products\FMMOS%20Utilization%20Reports\Class%20I%20Utilization%20of%20Producer%20Milk>, Accessed 07/20/2026.
- AGRICULTURAL MARKETING SERVICE. (2017c): “Class II Utilization of Producer Milk,”<https://mpr.datamart.ams.usda.gov/menu.do?path=\Products\FMMOS%20Utilization%20Reports\Class%20II%20Utilization%20of%20Producer%20Milk>, Accessed 07/20/2026.
- AGRICULTURAL MARKETING SERVICE. (2017d): “Class III Utilization of Producer Milk,”<https://mpr.datamart.ams.usda.gov/menu.do?path=\Products\FMMOS%20Utilization%20Reports\Class%20III%20Utilization%20of%20Producer%20Milk>, Accessed 07/20/2026.

- AGRICULTURAL MARKETING SERVICE. (2017e): “Class IV Utilization of Producer Milk,”<https://mpr.datamart.ams.usda.gov/menu.do?path=\Products\FMMOS\%20Utilization%20Reports\Class%20IV%20Utilization%20of%20Producer%20Milk>, Accessed 07/20/2026.
- AGRICULTURAL MARKETING SERVICE. (2018): “Class and Component Prices,”https://mpr.datamart.ams.usda.gov/menu.do?path=Products\Dairy\All%20Dairy\DY_CL102%20National%20Dairy%20Products%20Prices%20-%20Monthly, Accessed 08/13/2026.
- ALLAIRE, J., Y. XIE, C. DERVIEUX, J. MCPHERSON, J. LURASCHI, K. USHEY, A. ATKINS, ET AL. (2026): “rmarkdown: Dynamic Documents for R.”
- CALIFORNIA DEPARTMENT OF FOOD AND AGRICULTURE. (2017): California Dairy Statistics 2017, Technical report, Technical report.
- HESTER, J., H. WICKHAM, AND G. CSÁRDI. (2026): “fs: Cross-Platform File System Operations Based on `libuv`.”
- HLAVAC, M. (2022): “stargazer: Well-Formatted Regression and Summary Statistics Tables.”
- MÜLLER, K. (2025): “here: A Simpler Way to Find Your Files.”
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2015a): “Crop Production Annual Summary,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/crop-production-annual-summary>, Accessed 08/04/2024.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2015b): “Agricultural Prices,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/agricultural-prices>, Accessed 08/04/2024.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2015): “Cattle,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/cattle>, Accessed 08/08/2026.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2015c): “Cash Rents,” Washington, DC: U.S. Department of Agriculture-https://www.nass.usda.gov/Surveys/Guide_to_NASS_Surveys/Cash_Rents_by_County/, Accessed 04/05/2024.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2018c): “Crop Production Annual Summary,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/crop-production-annual-summary?date=2018-01>, Accessed 09/12/2022.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2018d): “Agricultural Prices,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/agricultural-prices?date=2018-01>, Accessed 09/12/2022.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2018a): “Milk Production,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/milk-production?date=2018-02>, Accessed 09/13/2022.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2018b): “Dairy Products Annual Summary,” Washington, DC: U.S. Department of Agriculture<https://esmis.nal.usda.gov/publication/dairy-products-annual-summary?date=2018-04>, Accessed 09/14/2022.
- NATIONAL AGRICULTURAL STATISTICS SERVICE. (2019): “2017 Census of Agriculture,” Washington, DC: U.S. Department of Agriculture<https://www.nass.usda.gov/Publications/AgCensus/2017/index.php>, Accessed 08/08/2026.
- NEW YORK DEPARTMENT OF AGRICULTURE AND MARKETS. (2017): Western New York Milk Marketing Area 2017 Annual Statistics Report, Technical report, Technical report.
- PEBESMA, E. (2018): “Simple Features for R: Standardized Support for Spatial Vector Data,” *The R Journal*, 10, 439–46.
- PEBESMA, E., AND R. BIVAND. (2023): *Spatial Data Science: With applications in R*, Chapman and Hall/CRC.

- POSIT TEAM. (2026): “RStudio: Integrated Development Environment for R,” Boston, MA: Posit Software, PBC <http://www.posit.co/>.
- R CORE TEAM. (2025): “R: A Language and Environment for Statistical Computing.”
- RODRIGUEZ-SANCHEZ, F., AND C. P. JACKSON. (2025): “grateful: Facilitate Citation of R Packages,”
- USHEY, K., AND H. WICKHAM. (2026): “renv: Project Environments,”
- WICKHAM, H. (2011): “The Split-Apply-Combine Strategy for Data Analysis,” *Journal of Statistical Software*, 40, 1–29.
- WICKHAM, H., M. AVERICK, J. BRYAN, W. CHANG, L. D. MCGOWAN, R. FRANÇOIS, G. GROLEMUND, ET AL. (2019): “Welcome to the tidyverse,” *Journal of Open Source Software*, 4, 1686.
- XIE, Y. (2014): “knitr: A Comprehensive Tool for Reproducible Research in R,” Ed. by. Victoria Stodden, Friedrich Leisch, and Roger D. Peng *Implementing Reproducible Computational Research*, Chapman, Hall/CRC.
- XIE, Y. (2015): *Dynamic Documents with R and Knitr*, 2nd ed. Boca Raton, Florida: Chapman, Hall/CRC.
- XIE, Y. (2025): “knitr: A General-Purpose Package for Dynamic Report Generation in R,”
- XIE, Y., J. ALLAIRE, AND G. GROLEMUND. (2018): *R Markdown: The Definitive Guide*, Boca Raton, Florida: Chapman, Hall/CRC.
- XIE, Y., C. DERVIEUX, AND E. RIEDERER. (2020): *R Markdown Cookbook*, Boca Raton, Florida: Chapman, Hall/CRC.